

Beijing-Dublin International College



AUTUMN TRIMESTER FINAL EXAMINATION - (2025/2026)

School of Computer Science

COMP3038J Multi-Paradigm Programming

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Time Allowed: 60 minutes

Instructions for Candidates:

Answer **All** questions.

BJUT Student ID:_____ **UCD Student ID:**_____

I have read and clearly understand the Examination Rules of both Beijing University of Technology and University College Dublin. I am aware of the Punishment for Violating the Rules of Beijing University of Technology and/or University College Dublin. I hereby promise to abide by the relevant rules and regulations by not giving or receiving any help during the exam. If caught violating the rules, I accept the punishment thereof.

Honesty Pledge:_____ **(Signature)**

Instructions for Invigilators

Non-programmable calculators are permitted.
No rough-work paper is to be provided for candidates.

Question 1:

a. Scala is a statically and strongly typed programming language, but it does not always require us specify the type of a variable. Explain why. When does the type of a variable need to be explicitly specified?

Give two examples of the code to assign the value 67 to a variable named `number`. The first example should not include an explicit type definition, while the second example should. **(10%)**

b. Define a method named `example`. This function should take three parameters of the following types and return an int value. Parameters:

- A function that takes two ints as parameters and returns a double (a)
- An int value (b)
- A double value

This function should call the function passed as the first parameter using the second parameter (b) and the value 12 as parameters. The result of this should then be added to the third parameter and returned from the function. **(10%)**

c. Scala has multiple different string interpolators that can be used. For two of the interpolators that we learned about in class provide the following information:

- The character or characters that must be put before a string to use it
- An explanation of what the interpolator is used for
- A code example showing the use of the interpolator

(10%)

d. Generic type parameters in Scala can be *invariant*, *covariant*, or *contravariant*, explain the difference between *invariant* and *covariant*. What is required in order to specify a covariant type parameter? Give an example of a generic type definition for *covariant* and *contravariant* type parameters. **(10%)**

e. Explain in the difference between imperative and declarative programming. Describe two possible benefits to code that is written declaratively. **(10%)**

(Question Total 50%)

Question 2:

a. List four of the core principles of functional programming. **(10%)**

b. Class variables and methods (commonly called static methods and variables) cannot be defined in Scala classes. How can this functionality be implemented in Scala? Give an example in Scala defining a class variable named `number` containing the constant 93 and a class method named `double` that takes a single int a parameter and returns the value multiplied by 2. **(10%)**

c. 1) Given a list of `Int` values named `numbers`, write an expression that will return a list containing only the numbers from the list that are divisible by 3.

   2) Given a list of `Double` values named `floats`, write an expression that will return a list containing these numbers where each has been multiplied by 3.

   3) Given a list of `Int` values named `nums`, write an expression that will return a list containing only the numbers less than 10, however these numbers should be modified such that each has been multiplied by 5.

   4) Given a list of `Int` values named `vals`, write an expression that will return a list containing these values multiplied by 5, but only where the result of the multiplication is greater than 5000.

   5) Given a list of `Person` values called `people`, write an expression that will return a list of String values representing the name (instance variable) of the person objects in the list.

   (10%)

d. What does it mean for a data type to be a monoid (or be monoidic)? Which higher order function makes good use of monodic types? **(10%)**

e. Data modelling is a core aspect of programming, create a "skinny domain model" for the university application described below.

   This basic modelling system contains teachers, students, classes, and exams. For each of these, the system remembers the following attributes:

   - **Teacher** - name, email, and subject.
   - **Class** - class code, class name, teacher and the year.
   - **Student** - name, email, classes they are registered to.
   - **Exam Result** - Grade, date of exam, student, and the class the exam was in.

   (10%)

(Question Total 50%)